

matics and instructions needed to complete the repair? Would he not save so much time? Also imagine, when a mechanic is repairing something and needs an expert's opinion, how would it be if the expert could see the object being repaired through the eyes of the mechanic and explain things in real-time?

How BMW has put all this to work

More than a year ago, BMW Group deployed one of the early examples of the metaverse, when the term was still not so well-known. They deployed an augmented reality (AR) application for vehicle concept and prototype engineering at their pilot plant in Munich. Vehicles and their components are visualised on a platform linked to the BMW Group's product data management system. Specialists can drag and drop CAD files of components from the web based database to the AR goggles, to reproduce the data in 3D and in their original size in a realistic environment. Users can modify the size and position of components with simple hand movements, literally playing around with the design. People at different locations around the world can team up to review designs and concepts together. This helps identify errors much earlier in the process, saving cost and time later. The company said in a press release that the AR application speeded up the process by around 12 months!

Last November, BMW announced the adoption of Nvidia's Omniverse to simulate every aspect of its manufacturing. The idea is for each of its 31 factories to have a fully-functioning, real-time digital twin capable of simulating production and finite scheduling, right down to work order instructions and robotics programming on the shop floor. This will help in improving product quality and output, while reducing manufacturing costs and unplanned downtime. It will en-



Fig. 3: Popular martial arts workout BodyCombat combines fitness with gaming in the metaverse (Courtesy: Les Mills)

able global teams to collaborate on designing and planning the factory in real time. With the digital twin of an entire factory, BMW engineers can quickly identify where and how each specific model's production sequence can be improved. It also gives BMW the flexibility to reconfigure its factories quickly to accommodate new model launches.

The company has also armed technicians with RealWear HMT-1 headsets, running TeamViewer, in 347 BMW centres across the US to reduce vehicle service times. Using voice commands, technicians can access and view technical documents on a high-resolution micro-display or remotely connect with BMW engineers for expert advice, all the time keeping their hands free to work! HMT-1's camera enables offsite engineers to see exactly what's in front of the local technician, so they can help them troubleshoot the issue in real time. The offsite expert can also markup live video feeds or image stills, which the mechanic can see in the headset's micro-display. According to BMW, vehicle repairs are now being completed 75% faster. And it also serves the purpose of on-the-job training of local technicians!

In a recent podcast, McKinsey senior expert Richard Ward ex-

plained, "BMW's new all-electric vehicle production line ran as a simulation for six months, building virtual cars on a one-to-one scale in the metaverse, before they actually did the final layout for the factory. And in the process of those six months, they changed the design about 30% from the original. They have not said publicly how much more efficient it was, but they did say that about 30% of what they thought was the world's best factory on day one of the simulation had to change in the process. These are people who build new factories daily, and they still found that level of learning from the simulation. So, why not practice first in the metaverse, where it's low-cost, you can do things in infinite ways, and you can make the impossible happen? Those types of things really offer people more efficiency and productivity."

Military training: Can't afford to learn on the job!

Governments across the world are experimenting with how the metaverse can be used for military training and strategising.

One of the oft-spoken-about examples is that of flight simulators used to train pilots. And now, real-time information overlay using AR headsets is being used to enable